

22

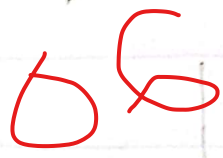
CSE 420 Final

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ID : 20201161

SEC : 08.

Input: record { float [2][3] a; int b; } P;



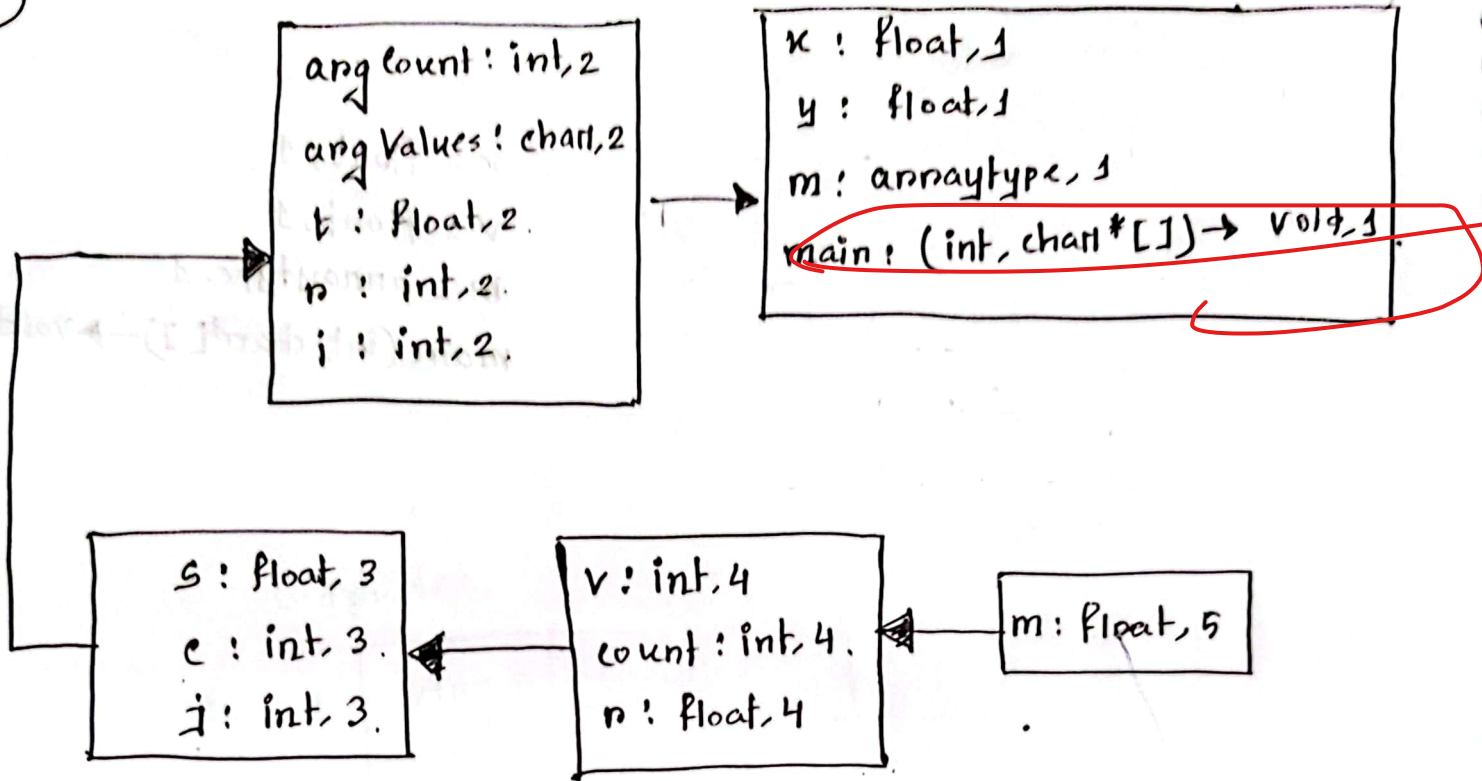
Symbol Table

Var	Type	Offset
a	array (2, array (3, float))	0
b	int	48
p	int	52

int = 4 bytes
array (2, array (3, float))
array (3, float)
float = 4 bytes
array (2, array (3, float))
array (3, float)
float = 4 bytes

float = 4 bytes
array (3, float)
array (2, array (3, float))
array (3, float)
float = 4 bytes

2

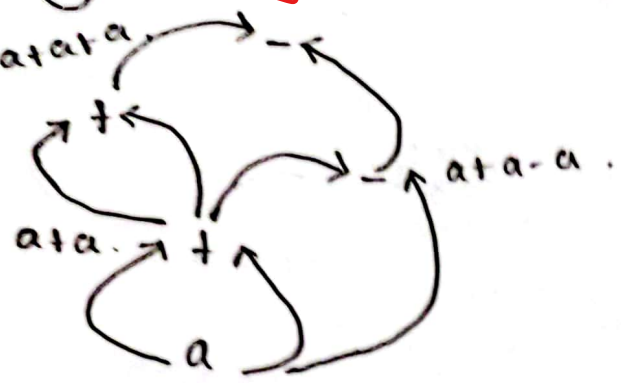


08

③
a+a+a-a

$$a+a+a-((a+a)-a)$$

The three address code.



- 0 $r_1 = a + a$
- 1 $r_2 = r_1 - a$
- 2 $r_3 = r_1 + r_1$
- 3 $r_4 = r_3 - r_2$

Quadruples.

Statements

Op	Arg 1	Arg 2	Result
+	a	a	r_1
-	r_1	a	r_2
+	r_1	r_1	r_3
-	r_3	r_2	r_4

Op	Arg 1	Arg 2
+	a	a
-	(0)	a
+	(0)	(0)
-	2	1